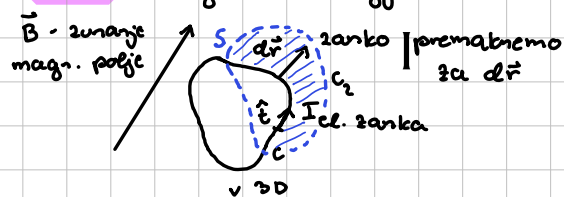


4.21 Magnetna energija

4.21.1 Magnetna energija v zunanem polju



Zanima nas, za koliko se je spremenila energija zanke (= koliko dela smo opravili).

Sila na zanko:
$$\mathbf{F} = \int_C I d\mathbf{l} \times \mathbf{B} = I \int_C (\mathbf{\hat{t}} \times \mathbf{B}) d\mathbf{l}$$

Če zanko premaknemo za $d\mathbf{r}$ opravimo delo:

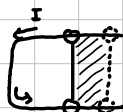
$$dA = -\mathbf{F} \cdot d\mathbf{r} = -I \int_C [(\mathbf{\hat{t}} \times \mathbf{B}) \cdot d\mathbf{r}] d\mathbf{l} = -I \int_C (d\mathbf{r} \times \mathbf{\hat{t}}) \cdot \mathbf{B} d\mathbf{l}$$

$d\mathbf{S}$ - diferencial površine, ki jo opiše zanka ob premiku.

Torej:

$$A = -I \int_S \mathbf{B} \cdot d\mathbf{S} = -I \Phi_m$$

Primer površine:



Zapis mag. energije z vekt. magn. poljem:

$$A = -I \int_S \mathbf{B} \cdot d\mathbf{S} = -I \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} =$$

vekt. mag. potoka

$$= -I \int_{C_2} \mathbf{A} \cdot d\mathbf{S} - (-I \int_{C_1} \mathbf{A} \cdot d\mathbf{r})$$

Posplošimo za gostoto toka $\mathbf{j}$:

$$A = - \int_V \mathbf{j} \cdot \mathbf{A} d^3\mathbf{r} + \int_V \mathbf{j} \cdot \mathbf{A} d^3\mathbf{r}$$

③ - po premiku ② - pred premikom

zunanji mag. polj. (odvisen od $\mathbf{j}$)

Torej → energija $\mathbf{j}$ v zunanjem mag. polju:

$$W = - \int_V \mathbf{j} \cdot \mathbf{A} d^3\mathbf{r}$$

4.21.2 Mag. energija kot funkcional toka

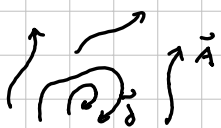
Iz Kirchhoffove enačbe (lahko) vemo: $\mathbf{\hat{A}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{j}'(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}'$

Potem lahko zapišemo:
$$W = \frac{\mu_0}{4\pi} \int_V \int_{V'} \frac{\mathbf{j}(\mathbf{r}) \cdot \mathbf{j}'(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r} d^3\mathbf{r}'$$

"kako eni tokovi ($\mathbf{j}$) delujejo na druge tokove ($\mathbf{j}$)"

4.21.3 Celotna magnetna energija

Zanima nas celotna energija polja/potenciala $\mathbf{A}$, ki ga ustvari gostota toka $\mathbf{j}$.



Uvedemo parameter: $\alpha: 0 \rightarrow 1$, pri čemer spremeni $\mathbf{A}: 0 \rightarrow \mathbf{\hat{A}}$

$\mathbf{j}$ ustvari $\mathbf{\hat{A}}$

- pri nekem α imamo $\mathbf{\hat{A}}$ in $\mathbf{\hat{A}} = \alpha \mathbf{j}$
- v $\mathbf{\hat{A}}$ dočemo $d\mathbf{j}$

$$\nabla^2(\mathbf{\hat{A}}\alpha) = -\mu_0(\mathbf{j}\alpha)$$

uporabimo linearnost Kirch. enačbe

$$dW = \int_V d\mathbf{j} \cdot \mathbf{A} d^3\mathbf{r} = \int_V -d\alpha \alpha \mathbf{j} \cdot \mathbf{A} d^3\mathbf{r}$$

$$W = - \int_0^1 \alpha d\alpha \int_V \mathbf{j} \cdot \mathbf{A} d^3\mathbf{r} \rightarrow \boxed{W = - \frac{1}{2} \int_V \mathbf{j} \cdot \mathbf{A} d^3\mathbf{r}} \quad \Delta$$

Vendar:

Prejšnji izraz ne upošteva, da je za vzpostavitve toka $\mathbf{j}$ potrebna energija (drugače kot v elektrostatiki, kjer so naboji statični).

Torej: $P = -\mathbf{U} \cdot \mathbf{I} = -I \int \mathbf{E} \cdot d\mathbf{r} = I \frac{\partial}{\partial t} \int \mathbf{B} \cdot d\mathbf{S} = \frac{\partial W}{\partial t}$ → zakaj smo I lahko nesli noter? Vzemimo.

(bo mogoče razločit naslednic)

→ moramo ustvarjati napetost, da tok teče, napetost ustvarjamo z indukcijo $\frac{\partial}{\partial t} \int \mathbf{B} \cdot d\mathbf{S}$

uporabimo ind. zakon:
$$\int \mathbf{E} \cdot d\mathbf{r} = - \frac{\partial}{\partial t} \int \mathbf{B} \cdot d\mathbf{S}$$

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$$

Preberemo: $W = I \int \mathbf{B} \cdot d\mathbf{S} = I \int \nabla \times \mathbf{A} \cdot d\mathbf{S} = I \int \mathbf{A} \cdot d\mathbf{r} = \int \mathbf{j} \cdot \mathbf{A} d^3\mathbf{r}$ (*)

Celotna energija torej: $W = \Delta + (*) = \frac{1}{2} \int \mathbf{j} \cdot \mathbf{A} d^3\mathbf{r} = W$

4.22 Gostota magnetne energije polja

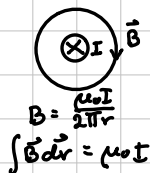
$$\nabla(\vec{B} \times \vec{A}) = \vec{A}(\nabla \times \vec{B}) - \vec{B}(\nabla \times \vec{A})$$

Celotno energijo želimo prepisati v odvisnost od $\vec{B}$:

$$W = \frac{1}{2} \int_V \vec{j} \cdot \vec{A} d^3\vec{r} = \frac{1}{2} \mu_0 \int_V (\nabla \times \vec{B}) \cdot \vec{A} d^3\vec{r} = \frac{1}{2\mu_0} \int_V \nabla \cdot (\vec{B} \times \vec{A}) d^3\vec{r} + \frac{1}{2\mu_0} \int_V \underbrace{\vec{B} \cdot (\nabla \times \vec{A})}_{\vec{B}} d^3\vec{r} =$$

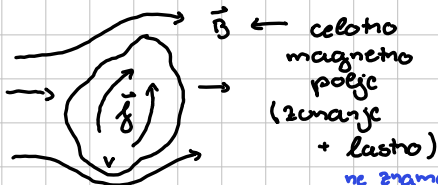
$$= \frac{1}{2\mu_0} \int_V B^2 d^3\vec{r} + \frac{1}{2\mu_0} \underbrace{\int_{\partial V} (\vec{B} \times \vec{A}) \cdot d\vec{S}}_{\propto \frac{1}{R}} \quad \begin{array}{l} \text{če je žica od } -\infty \text{ do } \infty \\ \text{to ne deluje} \rightarrow \text{v realnem} \\ \text{svetu ne obstajajo do žice} \end{array}$$

v limiti $R \rightarrow \infty$ je enako 0



Torej:
$$W = \frac{1}{2\mu_0} \int_V B^2 d^3\vec{r}$$

4.24 Sila kot funkcional magnetnega polja



Zanima nas, kakšna sila deluje na $\vec{j} \rightarrow$ porazdelitev toka.
Obračunavamo $\vec{j}$ v zunanjem magn. polju.

$$\vec{B} = \vec{B}_{\text{zun}} + \vec{B}_{\text{lasti}}$$

Velja:
$$\vec{F} = \int_V \vec{j} \times \vec{B} d^3\vec{r} = \frac{1}{\mu_0} \int_V (\nabla \times \vec{B}_{\text{lasti}}) \times \vec{B}_{\text{zun}} d^3\vec{r} =$$

$$\vec{F} = \frac{1}{\mu_0} \int_V (\nabla \times \vec{B}) \times \vec{B} d^3\vec{r}$$

$$\vec{B} \times (\nabla \times \vec{B}) = \frac{1}{2} \nabla B^2 - \nabla \cdot (\vec{B} \otimes \vec{B}) + \vec{B}(\nabla \cdot \vec{B})$$

$$\vec{F} = \frac{1}{\mu_0} \int_V \left[\nabla(\vec{B} \otimes \vec{B}) - \frac{1}{2} \nabla B^2 \right] d^3\vec{r} \rightarrow \boxed{\vec{F} = \frac{1}{\mu_0} \int_{\partial V} (\vec{B} \otimes \vec{B} - \frac{1}{2} B^2 \underline{\underline{I}}) d\vec{S}}$$

Integral poteka po površini "telesa"

ne znamo izmeriti $\rightarrow$ želimo formulo odvisno od celotnega $\vec{B}$
 $\nabla \times \vec{B} = \mu_0 \vec{j}$
 $\nabla \times \vec{B}_{\text{zun}} = 0$
 $\nabla \times \vec{B}_{\text{lasti}} = \mu_0 \vec{j}$
 $\vec{j} \times \vec{B}_{\text{lasti}} = 0$
 "sam sebe ne more odločiti"
 je lokaliziran, ne sme iti v $\infty \rightarrow$ rob mora biti končen

4.25 Tenzor napetosti magnetnega polja

Uvedemo tenzor napetosti mag. polja:

$$F_i = \oint_{\partial V} T_{ik} n_k dS \quad \text{normala na površino},$$

$$\text{kjer je } T_{ik} = \frac{1}{\mu_0} \left[B_i B_k - \frac{1}{2} B^2 \delta_{ik} \right]$$

Velja:

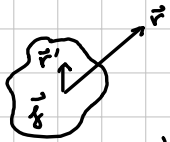
$$F_i = \int_{\partial V} T_{ik} n_k dS = \int_V \frac{\partial T_{ik}}{\partial x_k} d^3 \vec{r}$$

$\downarrow$ po površini telesa $\downarrow$ po volumnu telesa

$$g_i = \frac{\partial T_{ik}}{\partial x_k} \quad ; \quad g_i \rightarrow \text{volumska gostota sile}$$

4.27 Multipolni razvoj magn. polja

Štejimo nas $\vec{A}$ daleč stran od njegovega izvora...



$\vec{A} = ?$ "daleč stran"

$$\text{Velja: } \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{j}(\vec{r}')}{|\vec{r} - \vec{r}'|} d^3 \vec{r}'$$

$\leftarrow$ v praksi

Razvijemo: $|\vec{r}| \gg |\vec{r}'|$ dovolj faktor 5 do 10

$$\frac{1}{|\vec{r} - \vec{r}'|} = \frac{1}{r} - (\vec{r}' \cdot \nabla) \frac{1}{r} + \dots = \frac{1}{r} + \frac{\vec{r}' \cdot \vec{r}}{r^3} + \dots$$

Magn. potencial torej do II. reda lahko zapisemo kot:

$$\begin{aligned} \vec{A}(\vec{r}) &= \frac{\mu_0}{4\pi} \int_V \frac{\vec{j}(\vec{r}')}{r} d^3 \vec{r}' + \frac{\mu_0}{4\pi} \int_V \frac{\vec{j}(\vec{r}') (\vec{r}' \cdot \vec{r})}{r^3} d^3 \vec{r}' + \dots = \\ &= \frac{\mu_0}{4\pi r} \underbrace{\int_V \vec{j}(\vec{r}') d^3 \vec{r}'}_{\text{monopolni člen} \rightarrow \text{je vektor}} + \frac{\mu_0}{4\pi r^3} \underbrace{\int_V \vec{j}(\vec{r}') (\vec{r}' \cdot \vec{r}) d^3 \vec{r}'}_{\text{tudi dipolni člen je višjega reda}} + \dots \end{aligned}$$

Monopolni člen: $\int_V \vec{j}(\vec{r}') d^3 \vec{r}' = 0$, kar sledi iz pogoja, da so tokove zanke sklenjene.

Dipolni člen: $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi r^3} \int_V \vec{j}(\vec{r}') (\vec{r}' \cdot \vec{r}) d^3 \vec{r}'$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{r}}{r^3}, \quad \text{kjer } \vec{m}: \quad \vec{m} = \frac{1}{2} \int_V \vec{r}' \times \vec{j}(\vec{r}') d^3 \vec{r}' \leadsto \text{magnetni dipolni moment}$$

Magnetno polje magnetnega dipola:

$\leftarrow$ analogno el. polju el. dipola

$$\vec{B} = \nabla \times \vec{A} = \nabla \times \left(\frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{r}}{r^3} \right) = \frac{\mu_0}{4\pi} \frac{-r^2 \vec{m} + 3\vec{r}(\vec{m} \cdot \vec{r})}{r^5}$$

$\rightarrow$ magn. polje je analogno el. polju el. dipola

4.29 Amperova ekvivalenca



Izračunaj magn. dipolni moment tokovne zanke:

$$\vec{m} = \frac{1}{2} \int_V \vec{r}' \times \vec{j}(\vec{r}') d^3\vec{r}' = \frac{1}{2} \int \vec{r}' \times (I \vec{t}(\vec{r}')) dl = \frac{aI}{2} \hat{e}_z \int dl = I \pi a^2 \hat{e}_z$$

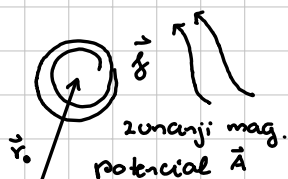
$S \rightarrow$ površina zanke

Cilindrična baza: $\hat{e}_r, \hat{e}_\phi, \hat{e}_z$

Tokovna zanka je ekvivalentna površina zanke magn. dipola.

4.30 Multipolni razvoj magnetne energije

Zanima nas magnetna energija multipolov v zunanjem magn. polju.



Gostota toka je lokalizirana okoli $\vec{r}_0$.

$$W = - \int_V \vec{j} \cdot \vec{A} d^3\vec{r}$$

$V \leftarrow$ po volumnu, kjer $\vec{j} \neq 0$

Razvijemo $\vec{A}$ okrog $\vec{r}_0$: $\vec{A}(\vec{r}) = \vec{A}(\vec{r}_0) + ((\vec{r} - \vec{r}_0) \cdot \nabla_0) \vec{A}(\vec{r}_0) + \dots$

$$W = - \int_V \vec{j} \cdot \vec{A}(\vec{r}_0) d^3\vec{r} - \int_V \vec{j} \cdot ((\vec{r} - \vec{r}_0) \cdot \nabla_0) \vec{A}(\vec{r}_0) d^3\vec{r} = - \vec{A}(\vec{r}_0) \cdot \underbrace{\int_V \vec{j} d^3\vec{r}}_{=0} - \underbrace{\int_V \vec{j} \cdot ((\vec{r} - \vec{r}_0) \cdot \nabla_0) \vec{A}(\vec{r}_0) d^3\vec{r}}_{\Delta} =$$

$$\Delta = \frac{\partial A_j(\vec{r}_0)}{\partial r_{0,i}}, \quad 0 = \int_V j_j (\vec{r} - \vec{r}_0)_i d^3\vec{r} = \int_V j_i (\vec{r} - \vec{r}_0)_j d^3\vec{r} \leadsto \text{glej naprej (*)}$$

koordinata (x, y, z)
uporabiš Gauss-Ostrogradski
razdelimo na polovico, za vsako polovico uporabimo svoj izraz

Simetrizacija tenzorja:

$$\int_V j_j (\vec{r} - \vec{r}_0)_i d^3\vec{r} = \frac{1}{2} \int_V [j_j (\vec{r} - \vec{r}_0)_i - j_i (\vec{r} - \vec{r}_0)_j] d^3\vec{r}$$

↑ nazaj v vektorski zapis

Potem:

$$W = - \frac{\partial A_j(\vec{r}_0)}{\partial r_{0,i}} \frac{1}{2} \int_V [j_j (\vec{r} - \vec{r}_0)_i - j_i (\vec{r} - \vec{r}_0)_j] d^3\vec{r} = - \frac{1}{2} \int_V [\vec{j} \cdot ((\vec{r} - \vec{r}_0) \cdot \nabla_0) \vec{A}(\vec{r}_0) - (\vec{r} - \vec{r}_0) \cdot (\vec{j} \cdot \nabla_0) \vec{A}(\vec{r}_0)] d^3\vec{r}$$

↑ primenimo vektore: $\vec{b} \cdot (\vec{a} \cdot \vec{c}) \vec{a} = \vec{a} (\vec{b} \cdot \vec{c})$

Uporabimo:

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c})$$

$$W = - \frac{1}{2} \int_V \underbrace{[(\vec{r} - \vec{r}_0) \times \vec{j}]}_{\vec{m}} \cdot \underbrace{[\nabla_0 \times \vec{A}(\vec{r}_0)]}_{\vec{B}(\vec{r}_0)} d^3\vec{r} \leadsto \boxed{W = - \vec{m} \cdot \vec{B}}$$

4.31 Sila in navor na dipol v zunanjem magn. polju

Sila:

$$dW = -\vec{F} \cdot d\vec{r} \sim \text{najhen premik dipola}$$

$$d(\vec{m} \cdot \vec{B}) = \underbrace{\nabla(\vec{m} \cdot \vec{B})}_{\text{arrow}} d\vec{r}, \quad \nabla(\vec{m} \cdot \vec{B}) = \vec{m} \times (\nabla \times \vec{B}) + (\vec{m} \cdot \nabla) \vec{B}$$

Sila na magnetni dipol: $\boxed{\vec{F} = (\vec{m} \cdot \nabla) \vec{B}}$

Navor: Velja: $dW = -\vec{M} d\vec{\varphi}$

$$dW = -d\vec{m} \cdot \vec{B} = -(\underbrace{d\vec{\varphi} \times \vec{m}}_{\substack{\uparrow \\ d\vec{m} = d\vec{\varphi} \times \vec{m}}}) \cdot \vec{B} = -d\vec{\varphi} \cdot (\vec{m} \times \vec{B})$$

Navor na magn. dipol:

$$\boxed{\vec{M} = \vec{m} \times \vec{B}}$$

⊛ Simetrizacija tenzorja → izrek Gauss - Ostrogradskega (2.72, p 31)

Tenzor spremeniš v tenor s preprostejšo strukturo.

Klasična formulacija: $\oint_{\partial V} \vec{A} \cdot d\vec{S} = \int_V \nabla \cdot \vec{A} d^3\vec{r}$

Uporabiš izrek G-O za tenzor III. reda $\vec{r} \otimes \vec{r} \otimes \vec{A}$

$$\oint_{\partial V} r_i r_j A_k \underbrace{n_k}_{\text{normala}} dS = \int_V [r_i A_j + r_j A_i + r_i r_j (\nabla \cdot \vec{A})] d^3\vec{r}$$

Uporabimo za $\vec{A} = \vec{j}_g$:

$$\underbrace{\oint_{\partial V} r_i r_j j_k n_k dS}_{=0 \text{ (ker zapobjameno z volumnom vse tokovne zanke)}} = \int_V [r_i j_j + r_j j_i + \underbrace{r_i r_j (\nabla \cdot \vec{j}_g)}_{=0 \text{ (ker tokovne zanke sklenjene)}}] d^3\vec{r}$$

lokalizirano in
bezizumsko pojele

Sledi: $\int_V r_i j_j d^3\vec{r} = - \int_V r_j j_i d^3\vec{r} \rightarrow \text{lanko narediš: } \int_V r_i j_j d^3\vec{r} = \frac{1}{2} \int_V [r_i j_j - r_j j_i] d^3\vec{r}$